

Nanditha Naik

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Education

University of Minnesota, MS in Electrical and Computer Engineering September 2023 - Present

Vishwakarma Institute of Technology, Pune, B.Tech Electronics and Telecommunication Engineering August 2018 - May 2022

• Deep Learning, Natural Language Processing, Statistical learning and inference, Optimization Theory

Technical Skills

Programming Languages: Python, C/C++, MATLAB

Tools & Frameworks: Git/GitHub, Linux, Docker, PyTorch, XCAL/XCAP, SQL, Tableau

Publications

- [1] [Accepted] Rostand A. K. Fezeu*, Jason Carpenter*, Eman Ramadan, Faaïq Bilal, Ziyang Wu, Nanditha Naik, Duncan Joly, Dehkontee Chea Cuppah, Zhi-Li Zhang. "A Peek into 5G NSA vs. SA Control Plane Performance", ACM HotMobile, 2024.
- [2] [Manuscript in Preparation] Nanditha Naik*, Rostand A. K. Fezeu*, Monaik Goshal, Dimitrios Koutsonikolas, Zhi-Li Zhang. "Characterizing and Predicting Handover Latency in 5G Networks via ML-Driven Signal Analytics".

Experience

Research Assistant UMN Computer Networking Lab, University of Minnesota August 2024 - Present

- Conducting 5G and ML research under Prof. Zhi-Li Zhang, focusing on scalable, QoS-aware, and secure wireless systems.
- Developing ML-based models for 5G handover prediction using large-scale XCAL mobility traces and RAN KPIs (PCI, RSRP, SCell events).
- Engineered end-to-end pipeline for preprocessing, feature extraction, and model training to enhance mobility management in 5G networks.
- Analyzed 220+ hours of 5G SA/NSA traces to identify registration and RRC session setup delays per 3GPP specifications.
- Applied statistical modeling and AI techniques to evaluate QoS metrics (latency, jitter, loss)

Research and Development Intern Climate Change Research lab, Indian Institute of Tropical Meteorology July 2022 - July 2023

- Built low power ESP32-based system with sensors for real-time hydro-meteorological data collection and wireless transmission.
- Programmed MicroPython firmware with UART, I2C, SPI, and 1-Wire support; enabled BLE, GSM (SIM800L), and Wi-Fi communication.
- Integrated AT commands and SMTP protocol for GPRS data transmission, SMS alerts, and email-based fault notifications.
- Verified end-to-end system with SD logging, BLE tests, and solar-powered battery management using DC-DC converters.
- Applied LSTM models to CRNP data for predictive soil moisture analysis, enabling ML-based insights in embedded sensing systems.

Projects

Imitation Learning with Diffusion Policy for Sweeping Tasks

- Designed and implemented a robotic table-cleaning system using Diffusion Policy for learning from demonstrations.
- Collected 245 expert demonstrations via teleoperation with wrist-mounted RGB cameras and UR5 robot joint angle data.
- Trained a diffusion-based policy model to map sensory inputs to control actions for object manipulation tasks.
- Achieved high generalization performance across varied object positions and attributes, validating robustness of learned policy.

Fraudulent Transaction Detection using Graph Attention Networks

- Developed and optimised GAT, GATv2 models in PyTorch Geometric for detecting fraud in the Elliptic Bitcoin dataset (203K+ nodes).
- Achieved 98% validation accuracy and 0.98 F1 score with dynamic attention-based GATv2, surpassing GCN benchmarks.
- Integrated multi-head attention, GELU activation, and dropout to boost noise resilience and model generalization.
- Built modular training pipeline with custom metrics logging and model checkpointing for streamlined experimentation.

Comparative Optimization of ANNs, LSTMs, and Transformer-based LLMs for Time-Series Forecasting

- Implemented and benchmarked ANNs, LSTMs, and transformer-based LLMs (e.g., GPT, Time-Series Transformer) for sequence prediction.
- Analyzed impact of optimizers (SGD, Adam, RMSprop, Armijo Rule) on model convergence, MAE/RMSE performance, and training efficiency.
- Applied training enhancements like learning rate scheduling, batch normalization, gradient clipping, and dropout for performance tuning.
- Evaluated trade-offs in training time, memory usage, and forecasting accuracy across models using real-world datasets from Kaggle.